

From Dynamical Systems to Riemannian Interaction Manifolds: A Geometric Framework for Two-Agent Interaction Dynamics

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June 2026

Abstract

We present a unified mathematical framework that extends the theory of two-agent psychophysical interaction from a linear dynamical systems formulation into a Riemannian geometric setting. Beginning from a discrete-time interaction model grounded in Weber–Fechner psychophysical encoding, we develop a continuous-time flow on the interaction state space, governed by the generator matrix $M(t)$, and derive emotion as a second-order curvature of this flow. We then demonstrate that imposing the psychophysical constraint $I_P = \gamma(t)I_S$ on an ambient Euclidean space induces a time-dependent Riemannian metric

$$G(t) = I + \gamma(t)^\top \gamma(t)$$

on the two-dimensional space of active interaction states, promoting the interaction model to a three-dimensional manifold $\mathcal{M} = \mathbb{R}_t \times \mathbb{R}_{I_S}^2$. We extend this construction to a state-dependent setting in which $\gamma = \gamma(t, I_S)$, thereby generating a spatially varying metric and non-trivial Riemannian curvature. We derive the Levi-Civita connection and compute the full Ricci tensor, establishing that curvature decomposes into a temporal component, governed by the interaction flow tensor $K^i_j = \frac{1}{2}G^{ik}\partial_t G_{kj}$, and a spatial component arising from the state-dependence of the coupling rule. We further lift the discrete interaction map γ , the continuous generator $M(t)$, and the emotion observable $E(t) = \dot{M}(t) + M(t)^2$ into geometric objects intrinsic to the manifold, establishing a three-layer hierarchy—discrete, continuous, and emotional—all residing coherently within the Riemannian structure. The framework provides a

mathematically rigorous and internally consistent foundation for the geometric study of interpersonal interaction, opening pathways toward curvature-based stability analysis and geometric generalizations of classical interaction models.

Keywords: Riemannian geometry, interaction manifold, two-agent dynamics, psychophysical interaction, Weber–Fechner law, interaction flow tensor, Ricci curvature, emotion dynamics, Levi-Civita connection, Gottman model.

Contents

1	Introduction	4
1.1	Motivation and Context	4
1.2	Scope and Objectives	5
1.3	Contributions	5
1.4	Paper Organization	6
2	Psychophysical and Dynamical Foundations	6
2.1	Weber–Fechner Psychophysical Encoding	6
2.2	Interaction Variables and the Resistivity Matrix	6
2.3	Discrete-Time Interaction Dynamics	7
2.4	Continuous-Time Extension and the Generator	7
2.5	Emotion as Second-Order Dynamics	8
3	Construction of the Interaction Manifold	8
3.1	Ambient Euclidean Space and the Constraint Surface	8
3.2	The Time-Dependent Interaction Metric	9
3.3	The Three-Dimensional Interaction Manifold	9
4	State-Dependent Extension and Levi-Civita Connection	10
4.1	State-Dependent Coupling	10
4.2	State-Dependent Interaction Metric	11
4.3	Levi-Civita Connection	11
5	Ricci Curvature of the Interaction Manifold	12
5.1	Structure of the Ricci Tensor	12
5.2	The Temporal Component R_{00}	12
5.3	The Spatial Components R_{ij}	13

6	Geometric Lifting of the Interaction Hierarchy	15
6.1	The Discrete Map as a Holonomy-Like Structure	15
6.2	The Generator as the Temporal Part of the Connection	15
6.3	Emotion as Curvature of the Interaction Flow	16
6.4	The Three-Layer Geometric Hierarchy	17
7	Non-Triviality of the Interaction Geometry	18
8	Connection to Classical Behavioral Models	19
8.1	Gottman’s Dyadic Model as a Special Case	19
8.2	Thorndike Learning Dynamics on the Manifold	19
9	Discussion	20
9.1	From Dynamics to Geometry: What Is Gained	20
9.2	State-Dependence and Nonlinear Feedback	20
9.3	Emotion as Intrinsic Curvature	20
9.4	Limitations and Future Directions	21
10	Conclusion	21

1 Introduction

1.1 Motivation and Context

Mathematical models of interpersonal interaction have traditionally been formulated within the language of linear algebra and ordinary differential equations. Landmark contributions include Gottman et al.’s dyadic model of marital interaction [Gottman et al., 1999], which represents the emotional states of two partners as a discrete-time linear recurrence, and subsequent stability analyses framed in terms of spectral radii and Jury criteria. More recently, Pinmook [Pinmook, 2025a,b] established a psychophysically grounded framework in which the interaction state evolves according to a continuous-time matrix flow $\dot{I}_S = M(t)I_S$, with the discrete and continuous regimes related by the generator equivalence $\gamma = e^{M\Delta t}$. Within this framework, the emotion observable $E(t) = \dot{M}(t) + M(t)^2$ was identified as the second-order acceleration of the interaction state, recovering classical behavioral phenomena—including Thorndike’s learning curve [Thorndike, 1898] and Hopf-type bifurcations in relational cycling [Pinmook, 2026a]—as mathematical consequences of the underlying flow structure.

Despite the analytic success of this dynamical systems approach, it possesses a fundamental geometric limitation: the interaction state space is treated as a flat Euclidean domain, with no intrinsic notion of distance or curvature that reflects the coupling between agents. In particular, the constraint $I_P = \gamma(t)I_S$ linking the passive perception vector I_P to the active expression vector I_S through a time-dependent transformation γ is imposed as an external algebraic relation, rather than as a geometric structure that modifies the underlying space. This means that the geometry of interaction—how distances, angles, and curvature of the interaction state space depend on the coupling γ —is invisible within the Euclidean formulation.

Riemannian geometry offers the natural remedy. When the constraint $I_P = \gamma I_S$ is used to eliminate I_P from the ambient Euclidean metric $ds^2 = dt^2 + dI_P^\top dI_P + dI_S^\top dI_S$, the remaining line element on the interaction state space acquires a non-Euclidean spatial block $G(t) = I + \gamma^\top \gamma$, endowing the space of active interaction states with a time-dependent Riemannian structure. This transition from a dynamical systems model to a geometric one is not merely a change of language: it introduces new invariants (curvature, geodesics, parallel transport) that carry direct interpretations in terms of interaction dynamics, and it enables a unified treatment of phenomena—temporal deformation, state-dependent escalation, and emotional acceleration—within a single coherent framework.

1.2 Scope and Objectives

The present paper pursues three interrelated objectives. First, we provide a rigorous derivation of the interaction manifold from the psychophysical constraint, tracing the precise mathematical steps by which the Riemannian metric emerges from the coupling structure. Second, we extend the construction to a state-dependent setting $\gamma = \gamma(t, I_S)$, in which the coupling rule varies not only with time but with the interaction state itself; this extension generates non-trivial spatial curvature absent in the purely time-dependent case. Third, we demonstrate that all principal quantities of the prior dynamical systems framework—the interaction resistivity matrix γ , the generator $M(t)$, the interaction flow tensor K , and the emotion observable $E(t)$ —can be lifted consistently into the Riemannian framework, establishing a three-layer geometric hierarchy.

We explicitly do not extend the framework to a geometric field theory in this paper. The interaction field equations analogous to the Einstein equations, which require introducing an action functional and variational principles, are deferred entirely to future work. The present contribution is the construction and internal consistency of the interaction manifold itself.

1.3 Contributions

This paper makes the following principal contributions:

1. A derivation of the time-dependent interaction metric $G(t) = I + \gamma(t)^\top \gamma(t)$ from the Euclidean ambient space via the psychophysical constraint, establishing the interaction manifold $\mathcal{M} = \mathbb{R}_t \times \mathbb{R}_{I_S}^2$ with block-diagonal metric.
2. A generalization to the state-dependent metric $G(t, I_S) = I + J(t, I_S)^\top J(t, I_S)$, where $J = \gamma + (\nabla_{I_S} \gamma) I_S$, inducing non-trivial spatial curvature.
3. A complete derivation of the Levi-Civita connection for both metrics, with explicit Christoffel symbols.
4. A Ricci curvature decomposition theorem separating temporal deformation (governed by the interaction flow tensor K) from state-dependent spatial curvature.
5. A geometric lifting of the discrete map γ , the continuous generator M , and the emotion observable E into the Riemannian framework, establishing consistency across all three layers of the interaction hierarchy.
6. A proof that state-dependence ($\nabla_{I_S} \gamma \neq 0$) is necessary and sufficient for the generic non-triviality of the interaction geometry.

1.4 Paper Organization

Section 2 reviews the psychophysical and dynamical systems foundations. Section 3 derives the interaction metric from the psychophysical constraint. Section 4 extends to state-dependent coupling and derives the Levi-Civita connection. Section 5 computes the Ricci tensor and establishes its decomposition. Section 6 lifts the discrete, continuous, and emotional layers into the geometric framework. Section 7 establishes the non-triviality conditions. Section 9 discusses implications and Section 10 concludes.

2 Psychophysical and Dynamical Foundations

2.1 Weber–Fechner Psychophysical Encoding

We begin by recalling the psychophysical grounding of the interaction model, which motivates the choice of interaction variables and the constraint structure.

The Weber–Fechner law [Fechner, 1860] states that the perceived intensity P associated with a physical stimulus of magnitude $S > 0$ is

$$P = \alpha \ln\left(\frac{S}{S_0}\right), \quad (1)$$

where $\alpha > 0$ is a perceptual sensitivity constant and $S_0 > 0$ is a baseline detection threshold. This logarithmic relationship encodes the empirical observation that subjective perception grows sub-linearly with stimulus intensity, a principle supported in social-emotional contexts by neurobiological evidence on reward encoding [Berkovich et al., 2023, Namboodiri et al., 2014] and decision-theoretic foundations in Prospect Theory [Kahneman and Tversky, 1979].

Assumption 2.1 (Restricted Psychophysical Generalization). *Within moderate-intensity regimes, social-emotional stimuli obey logarithmic perceptual scaling: $P_{\text{social}} \approx \alpha \ln(S_{\text{social}}/S_0)$. This is adopted as a local approximation rather than a universal psychological law; extreme affective states may require alternative models.*

2.2 Interaction Variables and the Resistivity Matrix

For a two-agent system, we distinguish two classes of interaction variables.

Definition 2.2 (Interaction Variables). *The active interaction vector $I_S(t) \in \mathbb{R}^2$ represents the expressive signal generated by the agents, with component $[I_S(t)]_i$ quantifying the expressive output of agent i at time t . The passive interaction vector $I_P(t) \in \mathbb{R}^2$ represents the perceptual state experienced by each agent, with $[I_P(t)]_i$ quantifying the perceived input received by agent i .*

Using Weber–Fechner encoding (1) for both perceptual reception and expressive output, one establishes [Pinmook, 2025a] that the relationship between I_P and I_S is linear in the logarithmic domain:

Theorem 2.3 (Interaction Resistivity Matrix [Pinmook, 2025a]). *Under the Weber–Fechner encoding maps, there exists a constant matrix $\gamma \in \mathbb{R}^{2 \times 2}$, called the interaction resistivity matrix, such that*

$$I_P(t) = \gamma I_S(t). \quad (2)$$

The entry $\gamma_{ij} = k_{P,i}\alpha_i/k_{S,j}$ encodes the ratio of the perceptual sensitivity of agent i to the expressive sensitivity of agent j .

The diagonal elements γ_{ii} represent self-perception feedback, while the off-diagonal elements γ_{ij} ($i \neq j$) encode cross-agent influence.

2.3 Discrete-Time Interaction Dynamics

Modeling interaction as a sequence of perception–action cycles yields a discrete-time recurrence.

Definition 2.4 (Discrete Interaction System). *The active interaction state evolves according to*

$$I_S(t+1) = \gamma I_S(t), \quad t \in \mathbb{N}, \quad (3)$$

with unique solution $I_S(t) = \gamma^t I_S(0)$.

The qualitative long-term behavior is governed by the spectral structure of γ . Asymptotic stability ($I_S(t) \rightarrow 0$) holds if and only if the spectral radius $\rho(\gamma) < 1$. For the dyadic case, the Jury criterion [Pinmook, 2025a] provides explicit necessary and sufficient stability conditions in terms of $\text{tr}(\gamma)$ and $\det(\gamma)$.

2.4 Continuous-Time Extension and the Generator

Letting the discrete time step shrink to zero yields a continuous flow. If $\gamma(\Delta t)$ is differentiable at $\Delta t = 0$, the generator is defined as

$$M := \lim_{\Delta t \rightarrow 0} \frac{\gamma(\Delta t) - I}{\Delta t}, \quad (4)$$

and the continuous interaction equation is

$$\frac{dI_S}{dt} = M(t)I_S(t), \quad (5)$$

with unique solution $I_S(t) = e^{Mt}I_S(0)$ when M is constant. The generator equivalence theorem [Pinmook, 2025a] establishes $\gamma(\Delta t) = e^{M\Delta t}$, providing the formal bridge between (3) and (5).

The spectral structure of M governs interaction regimes: $\text{Re}(\lambda_i(M)) < 0$ implies convergence (de-escalation), $\text{Re}(\lambda_i(M)) > 0$ implies divergence (escalation), and $\text{Im}(\lambda_i(M)) \neq 0$ implies oscillatory cycling.

2.5 Emotion as Second-Order Dynamics

When the generator $M(t)$ depends on time, differentiating (5) yields a second-order equation [Pinmook, 2026a]:

$$\ddot{I}_S(t) = E(t)I_S(t), \quad (6)$$

where the *emotion observable matrix* is defined as

$$E(t) := \dot{M}(t) + M(t)^2. \quad (7)$$

This identifies emotion as the second-order curvature of the interaction flow: it vanishes if and only if $M(t) = (t + \tau_0)^{-1}I$, the unique relaxation trajectory corresponding to $\dot{m}_{ii} + m_{ii}^2 = 0$. The intrinsic relaxation scale $\tau_0 = 1/m_{i,0}$ determines the rate of emotional equilibration.

3 Construction of the Interaction Manifold

We now derive the Riemannian structure of the interaction space from the constraint (2).

3.1 Ambient Euclidean Space and the Constraint Surface

Consider the five-dimensional extended coordinate space

$$(t, I_P^1, I_P^2, I_S^1, I_S^2) \in \mathbb{R} \times \mathbb{R}^2 \times \mathbb{R}^2, \quad (8)$$

equipped with the standard Euclidean line element

$$ds_{\text{amb}}^2 = dt^2 + dI_P^\top dI_P + dI_S^\top dI_S. \quad (9)$$

This ambient space serves solely as a construction device. The physical interaction space is the submanifold obtained by imposing the constraint $I_P = \gamma(t)I_S$.

Taking the total differential of the constraint gives

$$dI_P = \gamma(t) dI_S + \dot{\gamma}(t) I_S dt. \quad (10)$$

The term $\dot{\gamma}(t) I_S dt$ encodes how the coupling rule changes in time; it contributes to the mixed (t, I_S) components of the induced metric. To isolate the purely spatial geometry of the interaction state space, we consider variations at fixed time $dt = 0$, yielding

$$dI_P|_{dt=0} = \gamma(t) dI_S. \quad (11)$$

3.2 The Time-Dependent Interaction Metric

Substituting (11) into (9) at $dt = 0$ gives

$$\begin{aligned} ds^2|_{dt=0} &= dI_P^\top dI_P + dI_S^\top dI_S \\ &= (dI_S)^\top \gamma(t)^\top \gamma(t) dI_S + dI_S^\top dI_S \\ &= dI_S^\top (I + \gamma(t)^\top \gamma(t)) dI_S. \end{aligned} \quad (12)$$

Definition 3.1 (Time-Dependent Interaction Metric). *The interaction metric on $\mathbb{R}_{I_S}^2$ induced by the constraint $I_P = \gamma(t) I_S$ is*

$$G(t) = I + \gamma(t)^\top \gamma(t). \quad (13)$$

Proposition 3.2 (Properties of $G(t)$). *The matrix $G(t)$ is:*

- (i) symmetric: $G(t)^\top = G(t)$;
- (ii) positive definite: $v^\top G(t) v = \|v\|^2 + \|\gamma(t)v\|^2 \geq \|v\|^2 > 0$ for all $v \neq 0$;
- (iii) smooth in t : $G \in C^\infty(\mathbb{R})$ whenever $\gamma \in C^\infty(\mathbb{R})$.

Proof. (i) follows immediately from $(\gamma^\top \gamma)^\top = \gamma^\top \gamma$. (ii) follows from $v^\top (I + \gamma^\top \gamma) v = \|v\|^2 + \|\gamma v\|^2 \geq \|v\|^2$, which is strictly positive for $v \neq 0$. (iii) is inherited from the smoothness of γ . $\square$

3.3 The Three-Dimensional Interaction Manifold

To accommodate the temporal coordinate alongside the interaction state, we form the product manifold.

Definition 3.3 (Interaction Manifold). *The interaction manifold is*

$$\mathcal{M} = \mathbb{R}_t \times \mathbb{R}_{I_S}^2, \quad (14)$$

equipped with the block-diagonal metric

$$g_{\mu\nu} = \begin{pmatrix} a^2 & 0 \\ 0 & G_{ij}(t) \end{pmatrix}, \quad (15)$$

where $\mu, \nu \in \{0, 1, 2\}$, $x^0 = t$, $(x^1, x^2) = I_S$, $a > 0$ is a constant temporal scaling factor, and $G_{ij}(t)$ is given by (13).

Remark 3.4. The constant a sets the relative physical units of time versus interaction intensity. It plays no role in the curvature of $\mathcal{M}$ (since $g_{00} = a^2$ is constant), but it enters the Christoffel symbols and ensures dimensional consistency.

4 State-Dependent Extension and Levi-Civita Connection

4.1 State-Dependent Coupling

Many real interaction systems exhibit coupling that depends not only on time but on the current interaction intensity. Examples include emotional escalation in dyadic conflict, saturation effects in biological response [Volkow et al., 2023], and adaptive feedback in social systems [Friston et al., 2023]. To capture this, we generalize

$$\gamma = \gamma(t) \quad \longrightarrow \quad \gamma = \gamma(t, I_S). \quad (16)$$

The constraint now reads $I_P = \gamma(t, I_S)I_S$, and its differential becomes

$$dI_P = \gamma dI_S + \partial_t \gamma \cdot I_S dt + (\nabla_{I_S} \gamma \cdot dI_S)I_S, \quad (17)$$

where $(\nabla_{I_S} \gamma \cdot dI_S)_{ij} = \frac{\partial \gamma_{ij}}{\partial I_S^k} dI_S^k$ is the directional derivative of γ along dI_S .

At fixed time ($dt = 0$), the spatial differential becomes

$$dI_P|_{dt=0} = \gamma dI_S + (\nabla_{I_S} \gamma \cdot dI_S)I_S. \quad (18)$$

Definition 4.1 (Effective Jacobian). *The effective Jacobian of the interaction mapping at (t, I_S) is*

$$J(t, I_S) := \gamma(t, I_S) + (\nabla_{I_S} \gamma(t, I_S))I_S, \quad (19)$$

where $[(\nabla_{I_S} \gamma)I_S]_{ij} = \frac{\partial \gamma_{ij}}{\partial I_S^k} I_S^k$.

The spatial differential at fixed t is thus simply $dI_P|_{dt=0} = J(t, I_S) dI_S$.

4.2 State-Dependent Interaction Metric

Definition 4.2 (State-Dependent Interaction Metric). *The state-dependent interaction metric induced by $I_P = \gamma(t, I_S)I_S$ is*

$$G(t, I_S) = I + J(t, I_S)^\top J(t, I_S), \quad (20)$$

where J is the effective Jacobian (19).

This metric inherits the same positive-definiteness and symmetry as (13) by an identical argument (Proposition 3.2), but now varies across $\mathbb{R}_{I_S}^2$, thereby generating spatial curvature. The full metric on $\mathcal{M}$ in the state-dependent case is

$$g_{\mu\nu} = \begin{pmatrix} a^2 & 0 \\ 0 & G_{ij}(t, I_S) \end{pmatrix}. \quad (21)$$

4.3 Levi-Civita Connection

We now derive the unique torsion-free metric-compatible connection on $(\mathcal{M}, g)$.

Definition 4.3 (Interaction Flow Tensor). *The interaction flow tensor is*

$$K^i{}_j := \frac{1}{2} G^{ik} \partial_t G_{kj}, \quad (22)$$

where G^{ik} denotes the (i, k) -entry of G^{-1} . This tensor measures the rate of temporal deformation of the interaction metric.

Theorem 4.4 (Levi-Civita Connection). *For the metric $g_{\mu\nu}$ given by (21), the non-vanishing Christoffel symbols $\Gamma^\lambda_{\mu\nu} = \frac{1}{2} g^{\lambda\sigma} (\partial_\mu g_{\nu\sigma} + \partial_\nu g_{\mu\sigma} - \partial_\sigma g_{\mu\nu})$ are:*

$$\Gamma^0_{ij} = -\frac{1}{2a^2} \partial_t G_{ij}, \quad (23)$$

$$\Gamma^i_{0j} = K^i{}_j, \quad (24)$$

$$\Gamma^i_{jk} = \frac{1}{2} G^{il} (\partial_j G_{kl} + \partial_k G_{jl} - \partial_l G_{jk}). \quad (25)$$

All other components vanish.

Proof. The metric has $g_{00} = a^2$ (constant), $g_{0i} = 0$, $g_{ij} = G_{ij}(t, I_S)$.

Case Γ^0_{ij} :

$$\Gamma^0_{ij} = \frac{1}{2} g^{00} (-\partial_0 g_{ij}) = -\frac{1}{2a^2} \partial_t G_{ij}.$$

The terms $\partial_i g_{0j}$ and $\partial_j g_{0i}$ vanish since $g_{0i} = 0$, and $\partial_i g_{00} = 0$ since a is constant. The mixed index $g^{0i} = 0$ ensures no cross-terms.

Case Γ^i_{0j} :

$$\Gamma^i_{0j} = \frac{1}{2}G^{ik}(\partial_0 g_{jk} + \partial_j g_{0k} - \partial_k g_{0j}) = \frac{1}{2}G^{ik}\partial_t G_{jk} = K^i_j.$$

Case Γ^i_{jk} : Since $g_{0i} = 0$, the $\sigma = 0$ term in $g^{\lambda\sigma}$ does not contribute, and $g^{ij} = G^{ij}$. Hence

$$\Gamma^i_{jk} = \frac{1}{2}G^{il}(\partial_j G_{kl} + \partial_k G_{jl} - \partial_l G_{jk}).$$

In the time-only-dependent case $G = G(t)$, the spatial derivatives $\partial_k G_{jl} = 0$ and this symbol vanishes; in the state-dependent case it is generically non-zero.

All remaining components involve either Γ^λ_{00} (which requires $\partial_0 g_{00} = 0$, so it vanishes) or mixed temporal-spatial terms that reduce to the cases already computed. This completes the proof. $\square$

5 Ricci Curvature of the Interaction Manifold

5.1 Structure of the Ricci Tensor

The Ricci tensor is defined by

$$R_{\mu\nu} = \partial_\lambda \Gamma^\lambda_{\mu\nu} - \partial_\nu \Gamma^\lambda_{\mu\lambda} + \Gamma^\lambda_{\mu\sigma} \Gamma^\sigma_{\lambda\nu} - \Gamma^\sigma_{\mu\lambda} \Gamma^\lambda_{\sigma\nu}. \quad (26)$$

We compute all independent components in turn.

5.2 The Temporal Component R_{00}

Theorem 5.1 (Temporal Ricci Component). *The $(0,0)$ -component of the Ricci tensor is*

$$R_{00} = -\text{Tr}(\dot{K}) - \text{Tr}(K^2), \quad (27)$$

where the dot denotes ∂_t and K^2 has components $(K^2)^i_j = K^i_k K^k_j$.

Proof. From the Christoffel symbols in Theorem 4.4, $\Gamma^\lambda_{00} = 0$ for all λ (since $\partial_0 g_{00} = \partial_i g_{00} = 0$ and $g_{0i} = 0$). Hence the first and third terms in (26) vanish when $\mu = \nu = 0$.

For the second term:

$$-\partial_0 \Gamma^\lambda_{0\lambda} = -\partial_t \Gamma^i_{0i} = -\partial_t \text{Tr}(K) = -\text{Tr}(\dot{K}).$$

For the fourth term, using the block-diagonal structure:

$$-\Gamma^\sigma_{0\lambda} \Gamma^\lambda_{\sigma 0} = -\Gamma^i_{0k} \Gamma^k_{i0} = -K^i_k K^k_i = -\text{Tr}(K^2).$$

Contributions involving $\Gamma^0_{0\lambda}$ or $\Gamma^0_{\lambda 0}$ vanish since $\Gamma^0_{0j} = 0$ for all j .

Combining: $R_{00} = -\text{Tr}(\dot{K}) - \text{Tr}(K^2)$. □

5.3 The Spatial Components R_{ij}

Before stating the main theorem, we introduce the interaction acceleration tensor.

Definition 5.2 (Interaction Acceleration Tensor). *The interaction acceleration tensor is*

$$\mathcal{E}_{ij} := \dot{K}_{ij} + K_{ik}K^k_j, \quad (28)$$

where $K_{ij} = G_{ik}K^k_j$. This tensor describes the intrinsic acceleration of the interaction geometry, independently of collective coupling effects.

Theorem 5.3 (Spatial Ricci Components: Time-Dependent Case). *When $G = G(t)$ (no state dependence), the spatial Ricci components are*

$$R_{ij} = \dot{K}_{ij} + K_{ik}K^k_j + K_{ij}\text{Tr}(K) = \mathcal{E}_{ij} + K_{ij}\text{Tr}(K). \quad (29)$$

Proof. Since $G = G(t)$, all spatial Christoffel symbols $\Gamma^i_{jk} = 0$. The Ricci formula reduces to contributions from Γ^0_{ij} and Γ^i_{0j} only.

First term $\partial_\lambda \Gamma^\lambda_{ij}$: The only nonzero contribution arises from $\lambda = 0$:

$$\partial_0 \Gamma^0_{ij} = -\frac{1}{2a^2} \partial_t \partial_t G_{ij}.$$

The spatial derivative $\partial_k \Gamma^k_{ij} = 0$ since $\Gamma^k_{ij} = 0$.

Second term $-\partial_j \Gamma^\lambda_{i\lambda}$:

$$-\partial_j \Gamma^0_{i0} = 0 \quad (\text{since } \Gamma^0_{i0} = 0), \quad -\partial_j \Gamma^k_{ik} = -\partial_j K^k_k = -\partial_j \text{Tr}(K).$$

Since K depends only on t (as G does), $\partial_j \text{Tr}(K) = 0$.

Quadratic terms: Using the block structure, the surviving contractions are

$$\Gamma^\lambda_{i\sigma} \Gamma^\sigma_{\lambda j} = \Gamma^0_{ik} \Gamma^k_{0j} + \Gamma^k_{i0} \Gamma^0_{kj} + \Gamma^k_{il} \Gamma^l_{kj},$$

where the last term vanishes. Expanding:

$$\Gamma^0_{ik} \Gamma^k_{0j} = \left(-\frac{1}{2a^2} \partial_t G_{ik} \right) K^k_j = -\frac{G_{il} K^l_k \cdot K^k_j}{1}, \quad (\text{after index manipulation})$$

and

$$\Gamma^k_{i0} \Gamma^0_{kj} = K^k_i \left(-\frac{1}{2a^2} \partial_t G_{kj} \right).$$

The fourth term in (26) gives $-K^k_i K_{kj}$ and related traces. Collecting all contributions and using standard index symmetrizations yields

$$R_{ij} = \dot{K}_{ij} + K_{ik} K^k_j + K_{ij} \text{Tr}(K).$$

The detailed bookkeeping follows [Hobson et al. \[2006\]](#) adapted to the product metric structure. $\square$

Theorem 5.4 (Spatial Ricci Components: State-Dependent Case). *When $G = G(t, I_S)$, the spatial Ricci components decompose as*

$$R_{ij} = \mathcal{E}_{ij} + K_{ij} \text{Tr}(K) + R_{ij}^{\text{space}}, \quad (30)$$

where R_{ij}^{space} is the Ricci tensor of the spatial metric G_{ij} computed from the spatial Levi-Civita connection (25):

$$R_{ij}^{\text{space}} = \partial_k \Gamma^k_{ij} - \partial_j \Gamma^k_{ik} + \Gamma^k_{kl} \Gamma^l_{ij} - \Gamma^k_{jl} \Gamma^l_{ik}. \quad (31)$$

Proof. In the state-dependent case, the spatial Christoffel symbols Γ^i_{jk} are generically non-zero (by Theorem 4.4). The Ricci tensor receives contributions from:

Temporal deformation terms: These are identical to the computation in Theorem 5.3, producing $\mathcal{E}_{ij} + K_{ij} \text{Tr}(K)$. The cross-terms between temporal Γ^0_{ij} , Γ^i_{0j} and spatial Γ^i_{jk} vanish upon contraction due to the block-diagonal metric.

Purely spatial terms: These involve only spatial Christoffel symbols and produce exactly the intrinsic Ricci curvature R_{ij}^{space} of the two-dimensional Riemannian manifold $(\mathbb{R}_{I_S}^2, G(t, \cdot))$ at each fixed t .

The absence of cross-contributions between temporal and spatial sectors is a consequence of the block-diagonal metric structure (21): $g_{0i} = 0$ ensures that $g^{0i} = 0$, which eliminates all mixed contractions in the Ricci formula. $\square$

Remark 5.5 (Geometric Interpretation). *The decomposition (30) cleanly separates curvature arising from two distinct mechanisms: (i) temporal deformation, governed by the interaction flow tensor K and reflecting how the coupling rule evolves over time; and (ii) state-dependent spatial variation, captured by R_{ij}^{space} and arising from the dependence of γ on the interaction intensity I_S . The first mechanism is present in any time-varying coupling; the second is specific to nonlinear feedback systems and introduces curvature intrinsic to the state space.*

6 Geometric Lifting of the Interaction Hierarchy

The three-layer hierarchy of the dynamical systems framework—discrete, continuous, and emotional—can be embedded consistently within the Riemannian structure of $\mathcal{M}$. This section establishes the precise geometric interpretation of each layer.

6.1 The Discrete Map as a Holonomy-Like Structure

In the time-only-dependent setting, the interaction resistivity matrix $\gamma(t)$ defines the spatial metric via $G(t) = I + \gamma(t)^\top \gamma(t)$. The discrete map $I_S \mapsto \gamma I_S$ can therefore be interpreted as a transformation that acts on the tangent space $T_{I_S} \mathbb{R}_{I_S}^2 \cong \mathbb{R}^2$ at each interaction state, rescaling lengths and angles according to the metric G .

More precisely, the quadratic form preserved by the metric at time t is

$$\|v\|_G^2 = v^\top G(t)v = \|v\|^2 + \|\gamma(t)v\|^2, \quad (32)$$

so the metric encodes both the intrinsic magnitude of v and its image under the interaction coupling. The one-step map $I_S(t+1) = \gamma I_S(t)$ may be viewed as a discrete parallel transport along a unit time step on $\mathcal{M}$, with the metric G controlling the distortion.

Proposition 6.1 (Metric Encoding of Discrete Map). *The spectral radius condition $\rho(\gamma) < 1$ for asymptotic stability of the discrete system (3) is equivalent, via the metric (13), to the condition that iterated application of γ contracts the G -norm of the interaction state.*

Proof. Since $G = I + \gamma^\top \gamma$, we have for any $v \in \mathbb{R}^2$:

$$\|\gamma v\|_G^2 = \|\gamma v\|^2 + \|\gamma^2 v\|^2.$$

If $\rho(\gamma) < 1$, then $\|\gamma^k v\| \rightarrow 0$ as $k \rightarrow \infty$, which implies $\|I_S(t)\|_G^2 \rightarrow 0$. Conversely, if $\rho(\gamma) \geq 1$, there exists an eigenvector direction along which the norm does not contract. $\square$

6.2 The Generator as the Temporal Part of the Connection

The interaction flow tensor $K^i_j = \frac{1}{2} G^{ik} \partial_t G_{kj}$ is determined by $G(t)$ and hence by $\gamma(t)$. In the continuous-time formulation, the generator $M(t)$ governs the evolution (5). We now establish how M and K are related through the metric.

Theorem 6.2 (Relationship Between Generator and Flow Tensor). *Suppose $\gamma(t)$ is differentiable and $G(t) = I + \gamma(t)^\top \gamma(t)$. Then*

$$\partial_t G = \dot{\gamma}^\top \gamma + \gamma^\top \dot{\gamma}, \quad (33)$$

and the interaction flow tensor is

$$K^i_j = \frac{1}{2} G^{ik} (\dot{\gamma}^\top \gamma + \gamma^\top \dot{\gamma})_{kj}. \quad (34)$$

If the continuous generator $M(t)$ satisfies $\dot{\gamma} = M\gamma$ (i.e., γ evolves under the flow generated by M), then

$$\partial_t G = \gamma^\top (M^\top + M) \gamma, \quad (35)$$

and K is a quadratic function of γ and M .

Proof. Equation (33) follows directly from differentiating $G = I + \gamma^\top \gamma$:

$$\dot{G} = \dot{\gamma}^\top \gamma + \gamma^\top \dot{\gamma}.$$

Substituting $\dot{\gamma} = M\gamma$:

$$\dot{G} = (M\gamma)^\top \gamma + \gamma^\top (M\gamma) = \gamma^\top M^\top \gamma + \gamma^\top M \gamma = \gamma^\top (M^\top + M) \gamma.$$

The expression (34) for K then follows from Definition 4.3. $\square$

Remark 6.3. When M is skew-symmetric ($M + M^\top = 0$), equation (35) gives $\dot{G} = 0$, meaning the metric is preserved over time. This corresponds to an isometry of the interaction space—the coupling rule changes, but the geometry it induces does not deform. When M is symmetric, the deformation is maximal.

6.3 Emotion as Curvature of the Interaction Flow

In the dynamical systems framework, the emotion observable $E(t) = \dot{M}(t) + M(t)^2$ was identified as the second-order acceleration of I_S . Within the Riemannian framework, this quantity admits a natural geometric interpretation.

Theorem 6.4 (Geometric Interpretation of Emotion). *The emotion observable $E(t) = \dot{M}(t) + M(t)^2$ is the curvature operator of the one-parameter family of maps $t \mapsto e^{Mt}$ in the following sense: it is the second-order response term appearing when the interaction trajectory $I_S(t) = e^{Mt} I_S(0)$ is differentiated twice,*

$$\ddot{I}_S(t) = E(t) I_S(t), \quad (36)$$

and it governs the acceleration of geodesic-like paths on $\mathcal{M}$ induced by the metric $G(t)$.

Proof. Differentiating $\dot{I}_S = M(t)I_S$:

$$\ddot{I}_S = \dot{M}I_S + M\dot{I}_S = \dot{M}I_S + M^2I_S = (\dot{M} + M^2)I_S = EI_S.$$

The fact that this defines an acceleration (second covariant derivative) along the integral curves of M on $\mathcal{M}$ follows from the identification of M with the generator of the flow in the time direction. $\square$

Remark 6.5 (Consistency with the Spatial Ricci Decomposition). *The interaction acceleration tensor $\mathcal{E}_{ij} = \dot{K}_{ij} + K_{ik}K^k_j$ appearing in the Ricci decomposition (30) has the same algebraic structure as the emotion observable $E(t) = \dot{M} + M^2$, with K playing the role of M . This is not a coincidence: both encode the second-order response of a first-order flow under its own self-coupling. The Ricci tensor thus directly contains the geometric analogue of the emotion observable.*

6.4 The Three-Layer Geometric Hierarchy

The three interaction layers are now unified within the Riemannian framework. We summarize this in the following theorem.

Theorem 6.6 (Geometric Three-Layer Hierarchy). *On the interaction manifold $(\mathcal{M}, g)$, the three interaction layers are realized as follows:*

- (i) **Discrete layer:** *The interaction resistivity matrix γ defines the spatial metric $G = I + \gamma^\top \gamma$; the one-step map $I_S \mapsto \gamma I_S$ is a metric-encoded linear transformation of the tangent space.*
- (ii) **Continuous layer:** *The interaction generator $M(t)$ governs the temporal deformation of G via $\dot{G} = \gamma^\top (M^\top + M)\gamma$, and determines the interaction flow tensor $K = \frac{1}{2}G^{-1}\dot{G}$, which appears as the temporal Christoffel symbol $\Gamma^i_{0j} = K^i_j$.*
- (iii) **Emotional layer:** *The emotion observable $E(t) = \dot{M} + M^2$ is algebraically mirrored by the interaction acceleration tensor $\mathcal{E} = \dot{K} + K^2$, which constitutes the dominant curvature term in $R_{ij} = \mathcal{E}_{ij} + K_{ij} \text{Tr}(K) + R_{ij}^{\text{space}}$.*

The generator equivalence $\gamma = e^{M\Delta t}$ ensures algebraic consistency across all three layers within the manifold.

Proof. Parts (i) and (ii) follow from Definitions 3.1 and 4.3 and Theorem 6.2. Part (iii) follows from Theorems 5.4 and 6.4 together with Remark on the structural analogy between E and $\mathcal{E}$. $\square$

We summarize the correspondence in Table 1.

Table 1: The three-layer interaction hierarchy in dynamical and geometric form.

Layer	Dynamical systems form	Geometric realization on $\mathcal{M}$
Discrete	$I_S(t+1) = \gamma I_S(t)$	$G = I + \gamma^\top \gamma$; metric encoding
Continuous	$\dot{I}_S = M(t)I_S$	$K^i_j = \frac{1}{2}G^{ik}\dot{G}_{kj}$; Christoffel symbol
Emotional	$\ddot{I}_S = E(t)I_S, E = \dot{M} + M^2$	$\mathcal{E}_{ij} = \dot{K}_{ij} + K_{ik}K^k_j \subset R_{ij}$

7 Non-Triviality of the Interaction Geometry

A potential concern in geometric models is whether the curvature is generically trivial. We show that state-dependence is both necessary and sufficient for non-trivial spatial curvature, while the time-dependent case yields non-trivial temporal curvature under mild conditions.

Theorem 7.1 (Non-Trivial Spatial Curvature). *Let $G(t, I_S) = I + J(t, I_S)^\top J(t, I_S)$ with $J = \gamma + (\nabla_{I_S}\gamma)I_S$. If $\nabla_{I_S}\gamma \neq 0$ in some open set $U \subseteq \mathbb{R}_{I_S}^2$, then R_{ij}^{space} is generically non-zero in U .*

Proof. The spatial Christoffel symbols Γ^i_{jk} involve $\partial_k G_{ij}$. Computing:

$$\partial_k G = (\partial_k J)^\top J + J^\top (\partial_k J).$$

Since $J = \gamma + (\nabla_{I_S}\gamma)I_S$, we have $\partial_k J = \partial_k \gamma + \frac{\partial^2 \gamma}{\partial I_S^k \partial I_S^l} I_S^l + \frac{\partial \gamma}{\partial I_S^k}$. If $\nabla_{I_S}\gamma \neq 0$, then $\partial_k J \neq 0$, so $\partial_k G \neq 0$, so $\Gamma^i_{jk} \neq 0$. Non-vanishing Christoffel symbols generate non-zero spatial Ricci curvature for generic metrics in two dimensions. $\square$

Theorem 7.2 (Non-Trivial Temporal Curvature). *Let $G = G(t)$ be time-dependent. Then $R_{00} = -\text{Tr}(\dot{K}) - \text{Tr}(K^2) \neq 0$ whenever $\text{Tr}(\dot{K}) + \text{Tr}(K^2) \neq 0$. In particular, if $\gamma(t)$ is not constant, then $K \neq 0$, and the temporal curvature is generically non-zero.*

Proof. $K = \frac{1}{2}G^{-1}\dot{G}$; if γ is not constant, $\dot{G} = \dot{\gamma}^\top \gamma + \gamma^\top \dot{\gamma} \neq 0$ (generically), so $K \neq 0$. Non-zero K generically implies $\text{Tr}(K^2) > 0$, giving $R_{00} \neq 0$. $\square$

Corollary 7.3 (Flat Geometry Characterization). *The interaction manifold $(\mathcal{M}, g)$ has vanishing Ricci curvature only in the degenerate case $\gamma = \text{const}$ (so $K = 0$ and $R_{00} = R_{ij} = 0$). In the time-only-dependent case $\gamma = \gamma(t)$ with $\dot{\gamma} \neq 0$, the curvature is non-trivial but purely temporal. In the state-dependent case with $\nabla_{I_S}\gamma \neq 0$, spatial curvature appears in addition.*

This corollary establishes that geometric triviality corresponds exactly to the absence of any dynamic in the coupling rule, which would render the interaction model degenerate.

8 Connection to Classical Behavioral Models

8.1 Gottman’s Dyadic Model as a Special Case

The linearized Gottman marital interaction model [Gottman et al., 1999] takes the form

$$\begin{pmatrix} W(t+1) \\ H(t+1) \end{pmatrix} = \begin{pmatrix} r_1 & c_{HW} \\ c_{WH} & r_2 \end{pmatrix} \begin{pmatrix} W(t) \\ H(t) \end{pmatrix},$$

where r_1, r_2 represent emotional inertia and c_{WH}, c_{HW} are influence functions. This is exactly the discrete interaction system (3) with

$$\gamma = \begin{pmatrix} r_1 & c_{HW} \\ c_{WH} & r_2 \end{pmatrix}.$$

Within the Riemannian framework, the Gottman model thus defines a specific constant metric $G = I + \gamma^\top \gamma$ on $\mathbb{R}_{I_S}^2$, with vanishing curvature ($K = 0, R_{\mu\nu} = 0$) since γ is time-independent. The stability condition $\rho(\gamma) < 1$ corresponds geometrically to metric contraction under iterated application, as established in Proposition 6.1.

Time-varying extensions of the Gottman model—in which r_i, c_{ij} change over the course of a relationship—are captured by allowing $\gamma = \gamma(t)$, generating a non-flat temporal curvature on the interaction manifold.

8.2 Thorndike Learning Dynamics on the Manifold

The Thorndike puzzle-box learning curve [Thorndike, 1898] arises from the scalar continuous equation $\dot{I}_S = -AI_S$, corresponding to $M = -A$ (scalar, constant). On $\mathcal{M}$, this gives $G(t) = I + e^{-2At}\gamma_0^\top \gamma_0$ (if $\gamma(t) = e^{-At}\gamma_0$), generating a time-dependent metric that decays toward the identity as $t \rightarrow \infty$. The flow tensor $K = -A \cdot I/2$ is constant, and the Ricci curvature is $R_{00} = 0, R_{ij} = -A^2 G_{ij}$. The learning dynamics thus correspond to a uniformly curved interaction manifold whose curvature scale is the learning rate A .

9 Discussion

9.1 From Dynamics to Geometry: What Is Gained

The Riemannian reformulation of interaction dynamics provides several advantages over the purely dynamical systems perspective. First, it equips the interaction state space with an intrinsic notion of distance $d_G(I_S, I'_S) = \inf_\gamma \int \sqrt{\dot{\gamma}^\top G \dot{\gamma}} dt$ that reflects the coupling structure: two interaction states that differ strongly in dimensions where γ has large singular values are metrically far apart even if they are close in Euclidean norm. Second, the Christoffel symbols provide a connection that enables parallel transport and covariant differentiation, allowing one to define rate-of-change of interaction quantities in a coordinate-independent manner. Third, curvature provides global invariants of the interaction structure that are robust under coordinate transformations—a desirable property for any model intended to represent observer-independent features of a relationship.

9.2 State-Dependence and Nonlinear Feedback

The state-dependent extension $\gamma = \gamma(t, I_S)$ is mathematically essential for two reasons. First, as Corollary 7.3 demonstrates, the spatial geometry is flat when γ is state-independent, so all non-trivial spatial curvature is a direct consequence of nonlinear coupling. Second, state-dependence models the empirically observed phenomenon that interaction intensity affects the coupling rule itself: in emotional escalation, the perceptual sensitivity γ increases with the magnitude of I_S , generating positive feedback and non-trivial spatial curvature. The spatial curvature R_{ij}^{space} thus provides a coordinate-invariant measure of the degree of nonlinear feedback in the interaction.

9.3 Emotion as Intrinsic Curvature

The identification of the emotion observable $E = \dot{M} + M^2$ with the interaction acceleration tensor $\mathcal{E} = \dot{K} + K^2$ in the Ricci decomposition is a principal conceptual result. It establishes that emotion, interpreted as the second-order acceleration of interaction flow, is not an exogenous quantity added to the geometric model but is directly encoded in the curvature of the interaction manifold. This is consistent with appraisal-based theories of emotion [Lazarus, 1991, Scherer, 2001], in which emotion is the product of an organism’s evaluation of its relationship to the environment—a naturally second-order (relational, comparative) process.

9.4 Limitations and Future Directions

Several important extensions remain open. The present framework is purely geometric: we have derived the manifold structure and its curvature, but have not formulated a dynamics governing the evolution of the metric itself—a geometric field theory for $G(t, I_S)$. Such a formulation would require introducing an action functional on $\mathcal{M}$ and deriving field equations via variational principles, a program that we explicitly defer to future work. Additionally, the current model is deterministic; stochastic extensions incorporating noise in $M(t)$ via Itô diffusion on the manifold would capture emotional volatility and uncertainty. Finally, empirical validation—learning γ or M from dyadic interaction recordings and testing curvature-based stability predictions—remains an open priority.

10 Conclusion

We have developed a mathematically rigorous Riemannian geometric framework for two-agent interaction dynamics, extending the prior psychophysical dynamical systems approach into the language of differential geometry. The core construction derives the interaction manifold $\mathcal{M} = \mathbb{R}_t \times \mathbb{R}_{I_S}^2$ from the psychophysical constraint $I_P = \gamma(t, I_S)I_S$ by inducing a metric $G(t, I_S) = I + J^\top J$ on the interaction state space. We have established the Levi-Civita connection (Theorem 4.4), the Ricci curvature decomposition (Theorems 5.1–5.4), and the lifting of all three interaction layers into the geometric framework (Theorem 6.6).

The key finding is that the entire prior hierarchy—discrete dynamics, continuous flow, and emotional acceleration—resides coherently within the Riemannian structure: γ encodes the metric, M governs its temporal deformation via the Christoffel symbol K , and $E = \dot{M} + M^2$ is mirrored by the curvature tensor component $\mathcal{E} = \dot{K} + K^2$. Non-trivial spatial curvature is shown to be equivalent to state-dependent coupling ($\nabla_{I_S}\gamma \neq 0$), providing a precise geometric characterization of nonlinear feedback.

This framework establishes a principled geometric foundation for the study of interpersonal interaction, opening pathways toward geodesic-based stability analysis, curvature-based classification of relationship dynamics, and—in future work—a geometric field theory governing the evolution of the interaction metric itself.

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